

CHAPTER 4: SITUATION ANALYSIS – SRI LANKAN COCONUT INDUSTRY

Coconut industry in Sri Lanka has become one of the major plantation sectors that has given substantial impact to Sri Lankan Economy. As mentioned in an early chapter and since it contributes 1.3% of GDP, (table 4.1) indicates how important it is to the country. It has become further important once it was realized with further improvement, there is a possibility the industry could fall in to a positive direction. Unfortunately the industry at present is not on a very good stage due to many variables but history has proved positive thinking and correct management strategies which could lift the industry to a competitive stage.

As it realizes the importance, it is indeed a requirement to analyze and identify the correct position of the industry. It leads not only to identify the relevant issues pertaining to the industry but also to rectify suitable solutions to apply in a more strategical manner. Situation analysis could be done in micro and macro level and believe that they go in both directions as it involves domestic and international market globally.

The following sections elaborate the situation analysis in both aspects, which means in micro and macro levels. In micro point of view internal factors like strength and weakness and externally look like opportunities and threat to the industry. But if you look at in the macro point of view politically, economically, socially and technologically the key areas need to be analyzed in detail to identify the overall situation of the industry.

4.1 The Coconut Industry in the Current Context

As explained though Sri Lankan Coconut industry performs to a certain extend, there is a very large necessity with regard to actual requirements. According to statistics' (Table 4.1: Sri Lanka Coconut Industry -2009) though the country consumes 2.7bn nuts being the actual requirement locally and internationally is 3.8 -3.9bn nuts It shows that there is a very much clear and direct problem with regard to 1bn nut shortage which gives substantial impact to domestic market price increase and negative impact to export sales due to a comparative high price in the international market. The problem will fall to the worst position when cultivation reduces and cost production increases in the plantation and processing sectors.

However with the interfering of state sectors and starting to replant the coconut in a large scale in the North and East and also suggestions of providing fertilizer and other materials at subsidized rates give some motivation to push the industry in to a positive direction. But impact of those proposals becomes very little due to the common perception of the industry. It is not a very profitable venture and need to do some dramatic changes to become a profitable operation in a consistent manner.

Table 4.1: Sri Lanka Coconut Industry – In a nut shell: year 2009

Sri Lanka Coconut Industry 2009	
Area under Coconut	394,836
Total Area (Ha)	71,347
Estate Sector	323,489
Small Holdings	
Cultivation Subsidies	
Area Benefited (Ha)	10,816
Amount (Rs :)	234
Total Nut Consumption (Mn .Nuts)	2,762.18
House hold fresh nut consumption	1,815.96
Used by industry	946.22
Consumed locally	453.10
Exported	493.12
Value of Kernel products	
Total Sri Lanka Exports	813,911.00
Coconut products (Rs: Mn)	29,081.00
As % of total exports	3.57
Contribution to GDP (Constant factor cost Price 1996) %	
Total Agriculture	12
(Agriculture, Live Stock, Forestry & Fishing)	3.5
Total Plantation (Tea, Rubber & Coconut)	
Coconut Products	1.3

Sources: Department of Census & Statistics, Central Bank of Sri Lanka,
Coconut Cultivation Board & Coconut Development Authority, 2009



The statistic's revealed from the total agriculture sector in average 30% contribution comes from coconut based products. History says in the past decades GDP contribution was 3-5% from the coconut industry. The fact remains is how could it improve the industry to perform to that mark from the current position.

Currently the plantation sector (esp. in small holder's interest) is at a minimum level due to the high cost of production and is also struggling to make a profit. But at the same time there are few planters' who manage the plantations to cultivate a good crop in the average 7000-8000 per acre in a single pick. It shows if the interest is put in a positive manner & manages it properly that gives a good harvest in a coconut land. The processing sector presently faces a major problem with a nut shortage to make their manufacturing process to continue. At the same time though they identified the importance of technological application to improve the process, they always have a second thought with regard to the return of the investment. Marketing sector is struggling with a high price to competitiveness with other countries to gain the competitiveness for their high quality coconut products.

According to the industry sources cost of production of Nuts is the main problem of the coconut industry. The impact has become much greater by indicating 70% – 80% cost of production of any coconut based finished product decided by the nut price. In the current scenario especially in the plantation sector there are many direct and indirect variable substantial variances. This leads to fluctuate the cost of nut production.

In a coconut plantation it is a tedious process not only to plant the tree but also to grow the plant in different stages by giving necessary treatments and protecting it from many external factors. As a result it is essential to keep the land, soils and fertilizer treatments and also protecting from pest and weed to gain a good yield. It is also necessary to maintain the cost at a controllable position even after cultivating the crop (from the factors such as picking, collecting and transporting) to make it a profitable venture to sustain as an industry in the long term.

However cost of production of a nut is based not only on above mentioned expenses but also amount of nut cultivate for a pick. In Sri Lanka coconut estates number of picks has increased to 12 times (in average once a month) also gives some adverse effects to nut cultivation.

To count the cost production per nut total annual expense of the particular estate is divided by annual yield. It indicates average cost of production in that particular year for the relevant estate only. For the purpose of exploring how the cost of production is distributed among the variables, researcher has taken a particular estate which cultivates coconut trees on a large scale (1,575 acres) in a successful manner. In 2010 particular estate cost or production 10.28 which has been divided as follows:

<u>Cost Variable</u>	<u>Cost per nut (Rs :)</u>	<u>%</u>
• Picking and Collecting (Includes material & labour cost)	1.95	18.97
• Field work (Includes road construction/fencing Weeding/Fertilizing/moister)	2.41	23.44
• Converting cost for copra	0.18	1.75
• Dispatching (Collecting/transport/brokering)	0.72	7.00
• Other expenses (Rent/electricity/staff salaries/ Vehicle & official residence mainten:/ Labour allowances & admin: expen: etc:	5.02	48.84

Total cost of production per nut **10.28**

Sources: Annual expenditure of NLDB Maradavila coconut estate, Bingiriya; Year 2010

However the above figures are subjective in different estates under different conditions based on the size of coconut estate & location, possibility of utilizing the labour and cost of the labour, aging of the coconut trees & breed of the coconut tree, type of soil & other climatic & weather variables and also most importantly degree of applications & treatments are provided by the coconut planter to the relevant coconut estate.

When the situation is compared globally Sri Lanka's cost of production is comparatively very much in a higher position. This certainly directly affects the final selling price. It gives negative impact to reduce not only domestic consumption but also loose international market share under stiff completion conditions.

The problem will give further adverse affect when Sri Lanka is compared with other major coconut export countries number of coconut trees cultivated in an acre and also number of nuts that cab be picked in an acre. (Table 4.2) This again is performed by not only the cost factor but the basic criterias in the industry, such as number of trees and nut production comparatively in the declining stage. It indicates the importance of technological application is an indeed timely requirement to cultivate the highest of number of tress in the suitable lands and to get the maximum yield from those cultivated trees. With the assistance of right technology could improve the productivity effectiveness of the yield and simultaneously increase the cost efficiency. It provides substantial impact to reduce the cost of the production of the nuts. This will lead to sustain the industry especially in the international market in a consistent manner.

Table 4.2: Comparative position of Cost of production, Number of trees per acre & Nut cultivation with other main Coconut export countries

	Philippines	Indonesia	Malaysia	India	Sri Lanka*
Cost of Prod: / Nut (in SL Rs :)	7-8 per nut	8-9 per nut	10-12 per nut	9-10 Per nut	12-15 Per nut
No of trees Per acre (ave :)	75-80	70-75	65-70	65-70	64-75
Nut cultivation Per Pick/acre	8,500	7,000	4,500	6,000	2,850

* By considering total coconut cultivation & nut production in Sri Lanka: year 2010

Source: Coconut Statistical year book – APCC: 2010

When we look at the table 4.3 it provides a detailed idea about the coconut industry cost of production variances and different forms of coconut products behaviour along with export prices behaviour during the period of the last five years. Overall the most important point is total production came down drastically in year 2009 compared with 2008. That is a very crucial need to look closely since it was the first time the industry declined with in the period of five years. (Figure 4.1) This has resulted around one main crucial factor, which means a severe shortage of nut production is also indicated in the same table. At this time since the supply of coconut based products couldn't meet the demand and increases the cost of production effect to further increase nut price substantially. (Figure 4.3) This unavoidable price increases gave a very big impact not only to meet global marketing objectives under competitive environment but also to reduce domestic consumption.

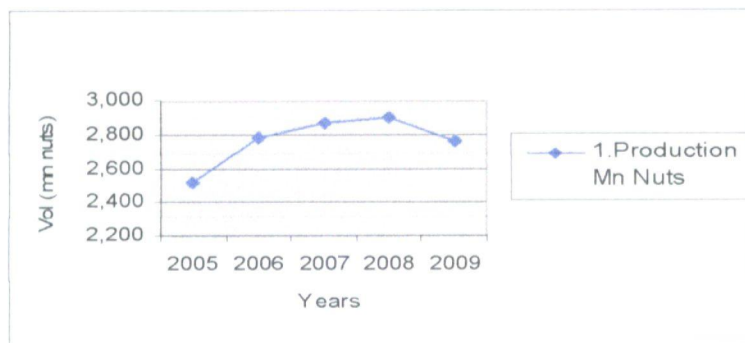
Table 4.3 : Key Indicators – Coconut Industry (year 2005 – 2009)

Item	unit	2005	2006	2007	2008	2009
1. Production	Mn: nuts	2,515	2,785	2,869	2,909	2,762
a. Desiccated Coconut	Mn: kgs	37	45	43	38	38
b. Coconut Oil	Mn: kgs	20	38	76	59	75
c. Copra	Mn: kgs	37	51	117	104	123
d. Poonac	Mn: kgs	10	18	42	30	38
2. Domestic nut consumption	Mn: nuts	2,047	2,222	2,451	2,437	2,269
3. Total Extent	'000 hec	395	395	395	395	395
4. Average Export Price (F.O.B.)	Rs/nut	16.29	24.26	27.56	28.96	22.92
5. Fertilizer issues	'000 MT.	33	47	25	29	25
6. Cost of production	Rs/nut		5	9	8.80	10.6
7. Replanting	ha	-	3,865	2,528	2,468	3,069
8. New planting	ha	-	4,101	2,955	3,683	4,356
9. Export earnings	Rs: mn	17,237	19,661	24,096	28,093	29,081
	US\$ mn	-	188	218	259	253
10. Kernel products	Rs: mn	7,719	9,142	11,270	13,667	11,974
	US\$ mn	-	87	102	126	104
11. Other products	Rs: mn	9,518	10,519	12,826	14,426	17,10
	US\$ mn	-	101	116	133	149

Source : Coconut Development Authority, 2009
 Department of Census & Statistics, 2009
 National Fertilizer Secretariat, 2009

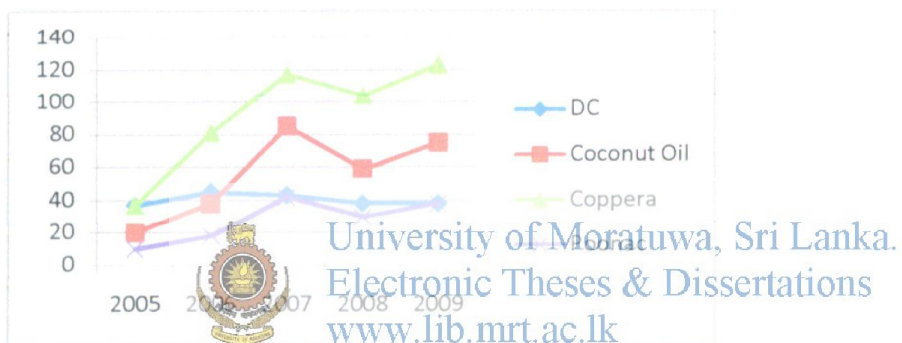
Even though replanting and new planting schemes have improved in the last few years that were not adequate at all to compensate and set off the negative impact that came from a high price. Reduction of cultivation due to poor income and wilt disease in the plantation sector and also the table emphasize reducing fertilizer issue & material price increases have come in a significant manner to increase the cost of production rapidly. It couldn't par with a new tree planting scheme which is also not happening to an expected phase & outcome. It is also given only after a period of time. This was the trend which goes towards in a negative direction that needs to take necessary controls as quickly as possible to turn into the right direction.

Figure 4.1: Coconut Production (Y: 2005-09)



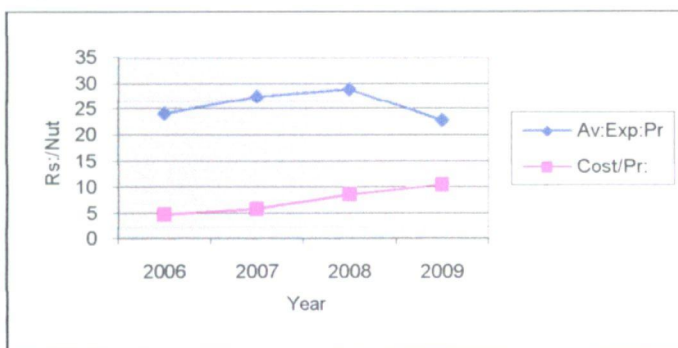
Source: Sri Lanka Coconut Statistics, 2009

Figure 4.2: DC, Coconut oil, Copera & Poonac Export Volumes (Y: 2005-09)



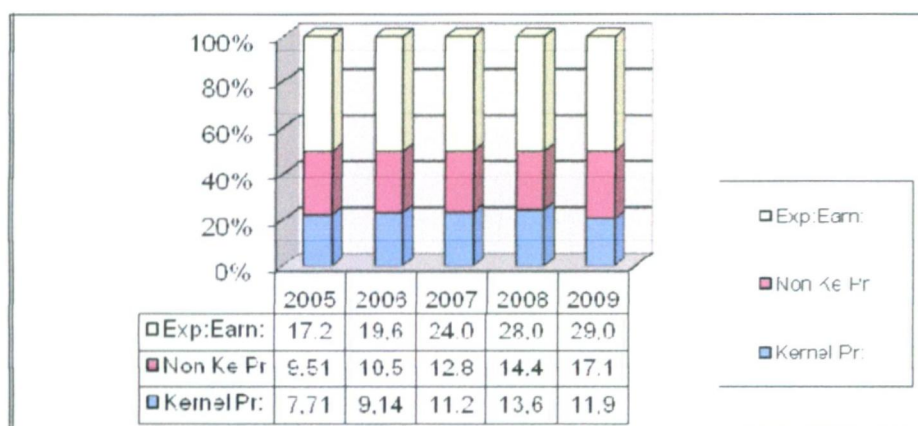
Source: Sri Lanka Coconut Statistics, 2009

Figure 4.3: Rs./Nut fluctuation of Ave: Export Price & Cost / Production (Y: 2006 -09)



Source: Sri Lanka Coconut Statistics, 2009

Fig 4.4: Total Coconut Export earning with Kernel & Non Kernel products forms
(Y: 2005 -09)



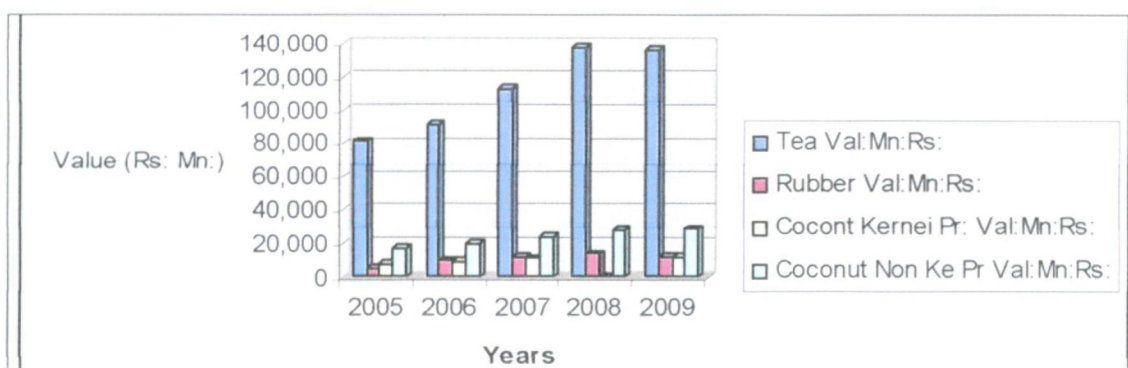
Source: Sri Lanka Coconut Statistics, 2009

Figure 4.5: Exports of Coconut Products: Year 2008 & 2009



Source: Sri Lanka Coconut Statistics, 2009

Figure 4.6: Value of Exports of Tea, Rubber & Coconut (2005-09)



Source: Sri Lanka Coconut Statistics, 2009

When you look at the industry in a macro point of view (compared with other plantations – Figure 4.6) coconut is comparatively at a lower stage According to table 4.4 it indicates the lowest production index is shown in the Coconut sector. The issue becomes worse when compared to the two years (2008 compared with 2007) % of rate of change shown as 1.4% as it is again the lowest in the index table.

Table 4.4: Agriculture Production Index (1997 – 2000 = 100)

Item	2007 (a)	2008 (b)	Rate of Change %
Agriculture & Fishing	114	124.4	10.1
1. Agriculture	111	125.7	10.2
1.1 Agriculture	113	122.9	10.7
Tea	106.3	110.6	4.1
Rubber	121.9	134	9.9
Coconut	103.3	104.8	1.4
Paddy	117.5	145.4	23.8
Other crops	111.5	121	8.5
1.2 Livestock	140.6	150.1	6.7
2. Fishing	106.6	116.9	9.6

(a) Revised

(b) Provisional

Source: Central Bank of Sri Lanka Report, 2009

In a macro point especially due to economical cost it leads to give tremendous impact to the plantation sector to reduce the crop. As a result it affects all the other areas such as coconut process and marketing sector to convert the industry to a profitable venture, before it is captured by other competitive countries or their subsidiary products. (Table 4.5). Political influence such as inappropriate duty structure & unconstructive management policies and decision makings affect to a greater extend (realized through interviews) to put the industry to unstable positions. Technological impact is an identified factor which needs to uplift the industry, but then again basic requirements (shortage of nuts) reflect indirectly to technological changes. Socially, people are well connected with the industry as it has a good relationship to their day today life style. But in the domestic market high prices and over all industry decline factors lead them to shift to another sector where it's viable in the long run.

Table 4.5: International Prices of Copra, Coconut oil & Other Subsidiary Products

Coconut Product Form	2008	2009
Copra (CIF .Europe)	816	476
Coconut Oil (CIF. Europe)	1,224	726
Palm Oil Crude (CIF .Europe)	949	683
Palm Kernel Oil (Malaysia CIF Rott)	1,130	702
Soya Bean Oil (Dutch / FOB Ex. Mill)	1,258	849
Sun Flower Oil (Ex-tank/EU. FOB	1,499	855
Rape Seed Oil (Dutch / FOB Ex. Mill)	1,329	859
Tallow (CIF Rotterdam)	885	681
Ground Nut Oil (CIF Rotterdam)	2,044	1,184

Source: Central Bank of Sri Lanka Report, 2009

4.2 Factors Effecting Coconut Industry

Factor analyzing is the most crucial strategy in any management practice to gain the effective outcome. As mentioned the above analysis has been done in micro and macro levels for the purpose of identifying the actual situation of the industry pertaining to all the directions. In micro analysis it has been done through SWOT analysis which will lead to go deeper to the industry and try to get in-depth knowledge. In the macro point of view, it does PEST analysis look in the wider scope and identify how those environmental factors affect to the industry which will give an impact to turn the industry in to positive or negative directions in a substantial manner.

4.2.1 Micro Point of View

Coconut is the industry which needs to be analyzed in detail especially by narrowing it down to a small process, as it has an impact not only to domestic life but also as a commercially viable industry. It helps to give a good picture and related issues of the industry's plantation sectors, which is owned by small scale owners and also processing units operated by rural people who are away from the main cities. In the micro level it is being done through SWOT analysis.



S – Strength

- The plantation which has in born strength suits not only country soil conditions but also weather patterns (rain falls)
- Cultivation suits most parts of the island which gives a good crop for the grower.
- Industry has been in the country and has given tremendous strength and proceeded in a positive direction due to high prominence.
- Industry which could turn round to a profitable path with minimum effort in a cost efficiency manner
- As a plant it has great value, which is satisfied with human basic needs, such as food and shelter, tempt to grow in a large scale to give commercial assistance.
- The industry, which is familiar with most of the people in domestic life boost to improve their consideration towards it.
- All parts of the tree have the ability of being converted to some consumable or industry related products, which could give commercial value
- As most of the people have got used to some traditional methods of processing to make finished products in a small and medium level, the knowledge of using methods are in a higher level.
- Industries become a greater strength to the country by giving contribution economically.
- In the manufacturing point of view most of the raw materials and processing methods could be found in the country it self.
- Most of the new practical innovative methods could be identified through industry, which will give positive net results
- Sri Lankan Coconut as a plant is eligible to make high standard products in bulk and value added form due to the high quality.

W – Weakness

- Lack of concentration in the plantation sector, from these who are away from the estates.
- Due to lack of concentration and poor management of estates lead to a severe nut shortage, which affects the total industry.
- Negative perception of the industry due to slow progress of the income (need minimum 5 years to get return)
- As the nature of the tree covers only 25 % of land area and balance 75% not useable to the coconut industry.
- Since the product is a very comparative product, price elasticity is very much greater against the demand.
- After some period it needs additional cost (fertilizer and chemicals) to maintain the estates in a required level.
- People who are in the traditional method of processing have some reluctance to get away from their use.
- Some times co product (nuts) gives similar impotents to different industries at the same time (Copra and desiccated coconuts)
- Social and economical implications (by not giving proper stands to the industry workmen) affect to recruit more skillful employers to the industry
- Industry doesn't have proper standard mechanism to collect nuts and send to the relevant international market or domestic supplier to gain the maximum benefits not only to the planter and manufacture but also to the final consumer.

O –Opportunities

- Politically it always tries to give more concentration, since it gives contribution to national economy.
- Technological applications, which help to improve productivity, could be implemented as mechanisms are not very complicated.
- Since the industry has been for a very long time, it has given an opportunity to penetrate new markets with the least minimum effort.

- Political stability of the country (end the war) leads to cultivate the plant in the North and East in Sri Lanka, it helps to overcome nut issue to a greater extent
- Since products could be used as recycled products, internationally it creates much demand to coconut based products
- Socially believed as the coconut gives prosperity it leads to positive attitudes to be protected as a tree
- Since the creative skill is in the higher stand, the average Sri Lankan has an opportunity to make different types of products, which could generate additional income to the country.

T – Threats

- Very high competition in the international market
- Threat from substitute products give greater threat in the domestic and international market
- Some of the processing mechanisms lead to pollute the environment
- High nut price leads to substantial disadvantage to country coconut based products
- Due to high price it gives an opportunity to enter competitive products to the domestic market (E.g.: Palm Oil)
- Economical adverse factors (recessions) lead to reduce country export volumes
- Some political differences with other countries and in appropriate agreements might lead to industry (E.g.: Vanspathi agreement with India)
- Inappropriate & imbalance taxes and duties affect to reduce industry volumes

4.2.2 Macro Point of view

Since the industry has gone beyond its borders and penetrates the global market, it is indeed a requirement to look at the industry wider to evaluate external factors to get a great knowledge about the environment. This will provide a long term impact to keep sustainability to take the right decisions at the right time. As a result industry PEST analysis has been done as follows:

P – Politically

- Political perception and their practical applications are in a very slow process which give adverse affects to progress
- Political stability in the country leads to give positive impressions from international buyers
- For the political benefit manipulating duties and taxes to protect one industry affect another industry negatively (Copra and DC)
- Politically need to support market oriented products and identify international new prospective markets
- Bureaucracy and unnecessary regulations reduce the efficiency of the process
- Politically not giving compensation to industry plays a greater role to sustain the coconut as an industry.

S - Socially

- Social link to the industry always gives optimistic outcome in all three processes
- Socially adamant about some processing methods and is difficult to change
- Since the social standards of the industry are not at expected level, this leads for people to get away from the industries that affect to divert it to a positive direction.
- For some social events parts of the tree are cut (flower of coconut) but replanting is a major issue
- Creative awareness is not in a required level to emphasis social importance to reduce consumption habits left away from coconut based products.

E – Economically

- Since it gives impact to the country's economy, domestic and international buying power affects country's GDP.
- Currency fluctuations and taxes and economical understanding with other countries (agreements) has affected to change the volumes
- Adverse effects of economical factors lead to increase the cost of production of the final product.
- Due to the nature of the product price elasticity plays a greater role and gives a negative impact in the international market
- Since the industry's nature as such is an opportunity cost at a minimum level leads to transfer substitute products.

T – Technologically

- Industry has not applied technological applications as expected to get the required outcome and necessity to do it accordingly to meet business objectives.
- Perception with regard to technology is not very positive. But to get the competitive edge it is more essential to improve industrial activities aligned with technology and increase the effectiveness to meet global demand.
- In the country technology is not being applied where it is necessary and gets away from the actual need and should meet a required situation change to think of innovative findings not only in a positive manner but also in a practical manner.
- Research findings always need to par with the latest methods (some occasions could get assistance from other counties) to benchmark our findings and convert into such a manner to use it appropriately.

4.3 Near Future Expectations

At the moment the plantation sector (especially with regard to main exports, namely Tea, Rubber & Coconut) are not doing that well, as performed in the past decades. There are many reasons behind that, mainly due to the economy opened in 1977. All restrictions pertaining to imports were removed and focused to drive to different directions or other industries which could generate more revenue to the country (E.g.: Apparel industry) However the majority of commercial groups are identified as a huge risk of these industries. Basically depend on uncontrollable external factors and think back to these traditional exports, which could generate the income at a comparably lower risk.

With regard to Sri Lankan Coconut industry, it is very much important to identify the issues in all three sectors (Plantation, Processing & Marketing) in depth and propose most economical practical solutions immediately to lift the industry to a better position. If necessary authorities responsibility to implement those applications with good intention (with short term and long term plans) changes it self to motivate others, who are not in the industry to reconsider & involve in a positive mindset. The fact is to prioritize the problems and provide necessary solutions accordingly (E.g.: start with basic issues like must shortage and reduce the cost of production)

If the solution arrives in some phase it will effect to improve our domestic market and most importantly their international market. As above mentioned in the SWOT & PEST analysis the country has an immense amount of positive factors to take coconut based products in many forms to the competitive market. The positive contributors are, it has been in the global market for a long period of time, ability of making & designing high quality value added products and global acceptance drive towards eco friendly reusable products etc: give high impact to expand the volumes to a very much greater extent. Since the political stability already arrived & also availability of huge unused land area which could be cultivated give greater boost growth. As those favorable factors naturally are gained by the country is to take those sales volume dimensions to a rapid growth in the long run.

4.3.1 Domestic Market Expectations

In the domestic market expectations are high, since the public always use & consume in a large scale compared with some other south Asian counties such as Philippines and Indonesia. If the industry falls into the correct directions, average Sri Lankan could buy coconut based products at a reasonable price (as some product prices are the highest in the world –E.g.: coconut oil) at any given time without any shortage. It will be an opportunity to an average Sri Lankans to use their own country products not only to consume it in many different forms, but also to align with modern fast life.

4.3.2 International Market Expectations

In a similar manner international expansions could be done if the necessities fall into a proper order. Most crucial factors like improvement of nut supply and make use of right technological applications to improve the quality and process efficiency. Also identification of new markets lead to take the industry to expected targets based on how efficient applications could be implemented.

Since the global attitudes gradually divert to natural consumer patterns and habits, these types of products and as an industry could play a substantial role to meet future international market expectations.

4.4 Summary of Situation Analysis

The chapter of the study described and detailed the situation (micro & macro point of view) with Sri Lankan Coconut industry, which is not favourable in many aspects. This will affect the domestic and international market and the effect will become a greater importance since it links with many ways to Sri Lankan economy such as GDP, balance of payment, employment and many indicators. When considering the micro point of view that did the analysis internally and externally through SWOT, it reveals though there are many strength and opportunities available and gained that we haven't capitalized those as expected to achieve the country objectives. At the same times some of the weaknesses and threats have arrived at a very crucial stage. They need to be attended as early as possible before the industry falls into a situation, where the country will have great difficulty to meet future aspirations. Not like the other industry it has become so vital, as invariably most of the people especially food consumption and many others used par with the tree and related products.

As a result protecting becomes further important to the industry domestically. Macro point of view the industry is struggling with not only meeting marketing expectations and up grading of technology not only to improve the productivity but also cost of production gives a huge impact and also act as a bottle neck to propose any innovative competitive solution to gain the edge. But overall when you look at the industry in a positive mindset, basic issues are countable and controllable. The need is to implement with the right action plan. That will motivate to turn around some macro factors such as economical & social issues, once they have seen the industry moves in the right direction. The next chapter would analyze the data gathered from customer survey to find out technology and marketing implication factors, which will drive the coconut industry to gain a competitive edge by adding value to Sri Lankan economy.